Problem Statement

- **Objective:** Build a 1D CNN model to detect irregular breathing patterns.
- **Dataset:** Labelled dataset of 8800 rows of 30-second windows, and an array of 960 signals per window.
- **Classification Labels are**:
    - Normal
    - Hypoapnea
    - Obstructive Apnea

Note: Other minority classes like the *Body event* and *Mixed Apnea* were merged to *Normal* and *Hypopnea* respectively, to avoid unnecessary classes and increase imbalance.

Model Architecture

A primary 8-layer model was selected as the baseline with L2 regularisation.

Loss: sparse_categorical_crossentropy
Optimizer: Adam
Epochs: 10
Batch Size: 32
Class Weight = "Balanced", Calculated using *computed class weight()*.

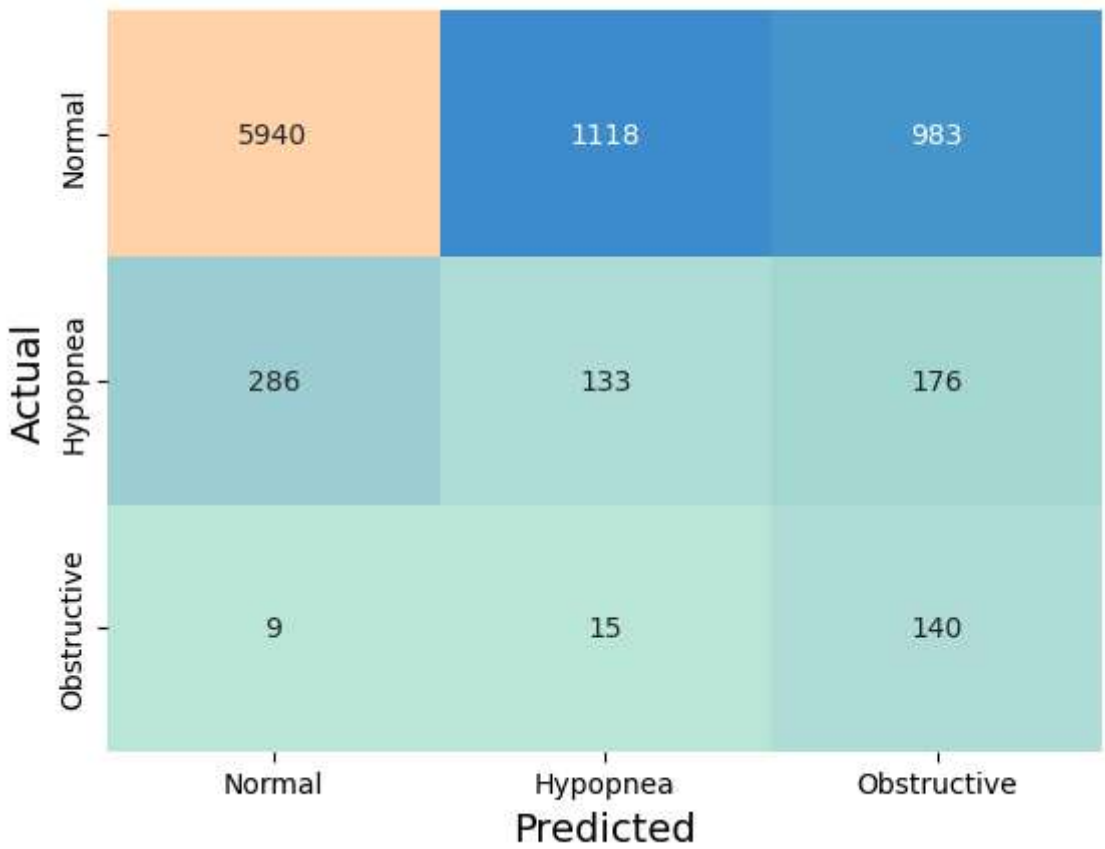
Model:

Layer	Configuration
Conv1D	32 filters, kernel size 5, ReLU
MaxPooling	pool size 2
Conv1D	64 filters, kernel size 5
MaxPooling	pool size 2
Flatten	—
Dense	64 neurons, L2 Regularisation (l2=0.007)

Dropout	0.4
Output	Softmax (3 classes)

Evaluation Metrics

Confusion Matrix :



Accuracy:

Average Accuracy Score = 0.706

It shows that the model was able to classify about 70% of the data correctly, which is not particularly impressive, since 90% of the windows are already labelled **Normal.**

Recall:

Average Recall Score = 0.60
Recall Per Class:

- Normal = $5940/(5940 + 1118 + 983)$ = **0.74**
- Hypopnea = $133/(286 + 133 + 176)$ = **0.22**
- Obstructive Apnea = $140/(9 + 15 + 140)$ = **0.85**

It shows that the model learnt well at capturing the actual Obstructive Apnea signals and Normal Breathing Signals, while it poorly captures Hypopnea signals.

Precision:

Average Precision Score = 0.39
Precision Per Class:

- Normal = $5940/(5940 + 286 + 9)$ = **0.95**
- Hypopnea = $133/(1118 + 133 + 15)$ = **0.11**
- Obstructive Apnea = $140/(983 + 176 + 140)$ = **0.11**

Precision of the Hypopnea and Obstructive Apnea is significantly poor; it is not able to differentiate between the sleeping disorder and normal out of the detected positive class.

Conclusion

Obstructive apnea is detected at a higher recall, indicating the model's capability to learn patterns associated with strong airflow suppression.
However, hypopnea classes are frequently misclassified as normal breathing, likely due to the subtle reduction in airflow amplitude that characterizes hypopnea events